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 Studierichting: Informatica
 Oefeningenreeks 6B nr. 4.2

$$x_1 = ?$$

$$\text{geg: } x_n = -1'', s_n = -45, v = -2$$

$$x_n = x_1 + (n - 1) \cdot v$$

$$s_n = \frac{x_1 + x_n}{2} \cdot n$$

$$\Leftrightarrow \begin{cases} -13 = x_1 + (n - 1) \cdot (-2) \\ -45 = \frac{x_1 - 13}{2} \cdot n \end{cases}$$

$$\Leftrightarrow \frac{-90}{x_1 - 13} = n \text{ als } x_1 \neq 0$$

$$\Leftrightarrow -13 = x_1 + \frac{180 + 2(x_1 - 13)}{x_1 - 13}$$

$$\Leftrightarrow -13 = \frac{x_1^2 - 13x_1 + 180 + 2x_1 - 26}{x_1 - 13}$$

$$\Leftrightarrow -13x_1 + 169 = x_1^2 - 11x_1 + 154$$

$$\Leftrightarrow x_1^2 + 2x_1 - 1 = 0$$

$$D = 64$$

$$\Leftarrow x_1 = -5 \text{ of } x_1 = 3$$

References